Large Steklov eigenvalues with a prescribed embedded boundary

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Abstract

For every smooth compact connected embedded boundary of positive dimension, the first positive Steklov eigenvalue of its smooth embedded fillings is unbounded, provided the ambient codimension is at least two and a filling exists. The boundary remains fixed pointwise, and the fillings carry their induced Euclidean metrics; volume and topology are unrestricted. We give a detailed proof using intrinsic large-eigenvalue constructions and relative smooth approximation of short embeddings. This follows the relative Nash–Kuiper mechanism announced by Girouard and Polymerakis, with the boundary and arbitrary-curve applications written out explicitly.

1 Introduction

The Steklov spectrum measures harmonic functions through their boundary values. On a smooth compact connected Riemannian manifold (M, g) with nonempty boundary, it is the spectrum of

$$\Delta_g u = 0 \quad \text{in } M, \qquad \partial_\nu u = \sigma u \quad \text{on } \partial M,$$

where ν is the outward unit normal. We write $0 = \sigma_0 < \sigma_1 \leq \sigma_2 \leq \cdots$. Fixing the boundary metric still leaves considerable freedom in the interior: in dimensions at least three, conformal changes can make σ_1 arbitrarily large while leaving that boundary metric unchanged [1]. Requiring the whole manifold to be embedded in one Euclidean space, with a prescribed embedded boundary, imposes an additional geometric constraint.

Colbois–Girouard–Gittins [2, Theorem 1.1] bound the Steklov eigenvalues of such submanifolds in terms of their fixed boundary and their volume. They ask in [2, Remark 1.7] whether the first positive eigenvalue can nevertheless tend to infinity when the boundary is fixed and the volume is free. Girouard, reporting joint work with Polymerakis, subsequently announced a relative Nash–Kuiper construction for transferring intrinsic examples to embedded ones, and described the case of a circle in $\mathbb{R}^3$ [5, p. 2075]. We follow that mechanism. The purpose of this note is to give a detailed relative approximation argument and to carry out its application to every admissible smooth embedded boundary, including arbitrary embedded curves.

Theorem 1.1. *Let $d \geq 1$ and $m \geq d + 2$ be integers. Let $\Sigma^d \subset \mathbb{R}^m$ be a smooth compact connected embedded submanifold without boundary which bounds at least one smooth compact connected embedded $(d+1)$-dimensional submanifold. For every $A > 0$ there is a smooth compact connected embedded filling $\Omega \subset \mathbb{R}^m$ such that*

$$\partial\Omega = \Sigma, \qquad \sigma_1(\Omega) > A,$$

where Ω has its induced Euclidean metric.

Thus, if $\mathcal{F}(\Sigma)$ denotes the entire class of these fillings, $\sup_{\Omega\in\mathcal{F}(\Sigma)}\sigma_1(\Omega)=\infty$. The theorem fixes the boundary as an ambient subset, rather than merely its intrinsic isometry class. It allows both volume and topology to vary. The volume dependence in the upper bound of [2] is consequently necessary in this setting.

An intrinsic large-eigenvalue metric is generally not the metric induced by the chosen embedding. We first match it to the induced metric on a shrinking boundary collar. We then enlarge it conformally in the interior, making the original embedding short: its induced metric is no larger than the target metric. This does not lower the Steklov eigenvalue. Finitely many local primitive-metric additions approximate the enlarged metric by the metric of a smooth embedding fixed on that collar. Uniform metric approximation controls the energy quotient and therefore retains the large eigenvalue. For surfaces, conformal enlargement leaves this energy unchanged; the intrinsic eigenvalues instead become large by allowing genus to increase, using the connected-boundary examples of [3].

Section 2 proves the relative approximation and spectral comparison. Section 3 applies them in the two filling dimensions $n=d+1\geq 3$ and $n=2$. All embeddings below take values in the original ambient $\mathbb{R}^m$, and σ_1 always means the first positive eigenvalue.

2 Relative approximation and the Steklov quotient

Throughout this section, M is a smooth compact connected manifold of dimension $n\geq 2$ with nonempty boundary. Metrics and maps are smooth up to the boundary. We use Riemannian volume densities, so orientability is not assumed. For a metric g, put

$$\mathcal{E}_g(u)=\int_M |\mathrm{d}u|_g^2\,\mathrm{d}V_g.$$

The first positive Steklov eigenvalue has the characterization

$$\sigma_1(M,g)=\inf_{\substack{u\in C^\infty(M),\ \int_{\partial M}u\,\mathrm{d}S_g=0\\ \int_{\partial M}u^2\,\mathrm{d}S_g>0}}\frac{\mathcal{E}_g(u)}{\int_{\partial M}u^2\,\mathrm{d}S_g}. \tag{1}$$

This follows from the smooth min–max formula in [1, Section 2, equation (2)]. Indeed, on the span of 1 and a mean-zero function the maximum quotient is the quotient of that function. Conversely, every two-dimensional subspace contains a nonzero function with zero boundary mean. A zero boundary norm is assigned infinite quotient. The sources [1, 3] number the zero eigenvalue as σ_1; their σ_2 is our σ_1.

2.1 Metric comparison

Lemma 2.1. *Suppose q and h have the same tangential boundary metric and*

$$(1-t)h\leq q\leq(1+t)h,\qquad 0<t<1.$$

Then

$$\sigma_1(M,q)\geq c_n(t)\sigma_1(M,h),\qquad c_n(t)=\frac{(1-t)^{n/2}}{1+t}. \tag{2}$$

If $h=\lambda^2 k$, where $\lambda\geq 1$ and $\lambda=1$ on ∂M, then

$$\sigma_1(M,h)\geq\sigma_1(M,k). \tag{3}$$

For $n=2$, equality holds in (3).

Proof. The inverse metrics and volume densities satisfy

$$q^{-1} \geq (1+t)^{-1} h^{-1}, \qquad \mathrm{d}V_q \geq (1-t)^{n/2}\, \mathrm{d}V_h.$$

Thus $\mathcal{E}_q(u) \geq c_n(t)\mathcal{E}_h(u)$. The boundary measures, mean-zero condition and denominators in (1) are identical, which proves (2). For the second assertion,

$$\mathcal{E}_{\lambda^2 k}(u) = \int_M \lambda^{n-2} |\mathrm{d}u|_k^2 \, \mathrm{d}V_k.$$

The boundary measure is again unchanged. The factor in the integrand is at least one for $n \geq 2$, and is exactly one for $n = 2$. □

2.2 Smooth embeddings fixed near the boundary

The approximation required below concerns the induced metric uniformly, and therefore only first derivatives of the embedding. We recall the local construction that also keeps a boundary collar fixed. A *primitive metric* on a coordinate chart is a tensor $a^2(\mathrm{d}\ell)^2$, where a is smooth with compact support in that chart and ℓ is linear in its coordinates.

For a symmetric tensor T and a positive metric h, write

$$\|T\|_h = \sup_{x \in M} \sup_{0 \neq v \in T_x M} \frac{|T_x(v,v)|}{h_x(v,v)}.$$

Lemma 2.2. *Let $f : M^n \to \mathbb{R}^m$ be a smooth embedding, $m > n$, and let $e = f^* g_{\mathrm{E}}$ be its induced Euclidean metric. Suppose*

$$h - e = \psi^2 B, \tag{4}$$

where B is a smooth positive definite metric and ψ is smooth with compact support in the interior of M. For every $\epsilon \in (0,1)$ there is a smooth embedding $F : M \to \mathbb{R}^m$, equal to f near the whole boundary, such that

$$(1-\epsilon)h \leq F^* g_{\mathrm{E}} \leq (1+\epsilon)h. \tag{5}$$

Proof. We use the primitive addition lemma of Eliashberg–Mishachev [4, Lemma 21.6.1, pp. 196–197]. In ambient dimension $m > n$, an embedding whose metric defect is one primitive metric can be approximated arbitrarily closely in C^0 by embeddings whose induced metrics approximate the target metric arbitrarily closely. We need the relative form of this finite construction; it follows from the parametric one-dimensional lemma used in that proof, as described next.

Work first in an interior coordinate box with primitive defect $a^2(\mathrm{d}\ell)^2$, and put $g = f^* g_{\mathrm{E}} + a^2(\mathrm{d}\ell)^2$. The source construction uses the leaves of $\ker \mathrm{d}\ell$ as transverse parameters and their g-normal curves as the longitudinal variable. A local Euclidean unit normal field ν to the image suffices. If v is the longitudinal g-unit vector, the formal longitudinal derivative

$$f_* v + a\, \mathrm{d}\ell(v)\nu$$

has Euclidean squared norm $e(v,v) + a^2 \mathrm{d}\ell(v)^2 = g(v,v) = 1$. It depends smoothly on the data and is the genuine derivative wherever $a = 0$. Lemma 17.5.1(c),(d) and the first remark after it in [4, pp. 161–162] keep the construction fixed near the transverse faces and the longitudinal endpoints when the data there are already genuine. Since a vanishes near the box boundary, the resulting perturbation is fixed near all its faces. Refining the support into smaller boxes, if needed, gives the same statement on an interior chart. This is an application of the relative parameter clauses to the proof of primitive addition, rather than an assumption of a smooth exactly isometric embedding.

The local output is an embedding by the same primitive addition lemma. To retain global injectivity, let K contain the compact support of one perturbation and choose a slightly larger chart U with $K \subset U$. For the current embedding, $f(K)$ and $f(M \setminus U)$ are disjoint compact sets, with positive Euclidean separation. Make the available C^0 error less than half that separation. Points in K cannot meet the complement of U; pairs in U are protected by the local embedding, and all other pairs are unchanged. The perturbation therefore extends by f to a global embedding. Coordinate boxes require only local normal fields, even when the global normal bundle is nontrivial.

We now decompose (4). The positive-cone argument in [4, Lemma 21.3.1, p. 193], applied to B on compact interior chart supports, and a smooth square partition of unity give

$$\psi^2 B = \sum_{j=1}^{N} p_j, \qquad p_j = a_j^2 (\mathrm{d}\ell_j)^2. \tag{6}$$

Here every a_j is smooth and supported compactly in one interior chart. For clarity, the square amplitudes can be obtained by first writing the positive matrix $B(x)$, locally, as a positive smooth combination of finitely many constant positive matrices. Choose these matrices as vertices of a small polytope containing the compact range under consideration, and split each vertex into squares of linear forms. The coefficients are strictly positive on smaller chart supports. Multiplication by ψ and the square-partition functions then gives $a_j = \psi \rho_j \sqrt{\beta_j}$ with smooth positive β_j. There is no square root of a vanishing arbitrary coefficient to regularize.

Starting from $f_0 = f$, at step j use the target

$$g_j = f_{j-1}^* g_{\mathrm{E}} + p_j.$$

The initial map at that step is short for g_j exactly. Choose the new metric error $E_j = f_j^* g_{\mathrm{E}} - g_j$ so that $\|E_j\|_h < \epsilon/N$, along with the C^0 tolerance just described. This is possible because $g_j \le C_j h$ for some finite C_j on compact M, so relative g_j error less than $\epsilon/(NC_j)$ suffices. The finite construction telescopes to

$$f_N^* g_{\mathrm{E}} = e + \sum_{j=1}^{N} p_j + \sum_{j=1}^{N} E_j = h + \sum_{j=1}^{N} E_j,$$

which proves (5).

All modifications are supported away from ∂M, and the intersection of their finitely many unchanged collars is still a boundary collar. The finite smooth constructions above give a smooth map. One may also use a C^1 formulation of the primitive lemma and then smooth relative to that collar with arbitrarily small C^1 error. Embeddings of a compact manifold are open in C^1: local graph estimates preserve injectivity on finitely many small charts, and compact separation handles the remaining pairs. Such smoothing preserves the embedding and changes its induced metric arbitrarily little. Reserving part of each error tolerance for smoothing gives the same conclusion. □

2.3 Passing from an intrinsic metric to an embedded filling

The intrinsic metric need not initially dominate the induced metric. A conformal enlargement can arrange this without decreasing the Steklov quotient. We first make it agree with the induced metric on a small collar.

Proposition 2.3. *Let $f : M^n \to \mathbb{R}^m$ be a smooth embedding, $n \geq 2$, $m > n$, and let k be any smooth metric whose tangential boundary restriction equals that of $e = f^* g_{\mathrm{E}}$. For every $\theta \in (0,1)$ there is a smooth embedding $F : M \to \mathbb{R}^m$, equal to f near ∂M, with*

$$\sigma_1(F(M)) \geq (1-\theta)\sigma_1(M,k). \tag{7}$$

The image carries its induced Euclidean metric.

Proof. Equality of tangential metrics does not match their normal components. Let C_e and C_k be collars obtained by flowing extensions of the inward unit normal fields of e and k. At the boundary, $C_k \circ C_e^{-1}$ fixes tangent vectors and sends the e-unit inward normal to the k-unit inward normal. It therefore pulls k back to e as a full tensor on $TM|_{\partial M}$.

This collar map extends to a boundary-fixing diffeomorphism of M. Indeed, interpolate the two inward transverse fields linearly. Their flows give an isotopy of collars on a common short interval, by compactness. On the image of a smaller collar its velocity in the isotopy parameter is a smooth time-dependent vector field, zero at boundary points. Cut it off inside a larger collar and integrate. Its flow agrees with the collar isotopy near the boundary and is the identity away from the larger collar. These flow and collar facts are the usual smooth constructions; see [6, Section 6.2]. Pulling back k by this diffeomorphism preserves its Steklov spectrum. Thus we may assume

$$k = e \quad \text{on } TM|_{\partial M}. \tag{8}$$

In an e-normal flow collar, let $t \geq 0$ be the inward coordinate. Choose $0 \leq \psi_\delta \leq 1$, smooth, equal to zero for $t \leq \delta$ and one for $t \geq 2\delta$, extended as one over the remainder of M. Set $\chi = \psi_\delta^2$ and

$$k_\delta = (1-\chi)e + \chi k.$$

By (8), the uniform e-norm of $k - e$ is $O(t)$ in the collar. Since $k_\delta - k$ is supported where $t < 2\delta$, $k_\delta \to k$ uniformly. For any fixed $\eta \in (0,1)$ we can choose $\delta > 0$ so that

$$(1-\eta)k \leq k_\delta \leq (1+\eta)k, \qquad \sigma_1(M,k_\delta) \geq c_n(\eta)\sigma_1(M,k).$$

The last inequality is lemma 2.1. This collar width is chosen separately for the given k.

Choose $c \in (0,1]$ such that $k \geq ce$ on M, and then $K > (1-c)/c$. Put

$$h = (1 + K\chi)k_\delta.$$

The conformal factor is at least one and equals one on the boundary collar. Thus $\sigma_1(M,h) \geq \sigma_1(M,k_\delta)$, including dimension two. Moreover,

$$\begin{aligned} h - e &= \chi B, \\ B &= Ke + (1+K\chi)(k-e) \geq [Kc - (1-c)]e > 0. \end{aligned}$$

The displayed formula defines a smooth positive B even at $\chi = 0$. Consequently $h - e = \psi_\delta^2 B$ has the form required by lemma 2.2.

Apply that lemma with tolerance ϵ. The induced metric $q = F^* g_{\mathrm{E}}$ satisfies (5) and has exactly the boundary metric of h. Hence

$$\sigma_1(F(M)) \geq c_n(\epsilon)\sigma_1(M,h) \geq c_n(\epsilon)c_n(\eta)\sigma_1(M,k).$$

For this fixed n, choose η, ϵ small enough that the product exceeds $1-\theta$. This proves (7). □

3 Proof of the prescribed-boundary theorem

Put $n = d + 1$. We use proposition 2.3 first on a fixed filling in dimensions $n \geq 3$, and then on a family of embedded surfaces when $n = 2$.

3.1 Filling dimension at least three

Choose any smooth compact connected embedded filling M of Σ and let e be its induced Euclidean metric. Colbois–El Soufi–Girouard [1, Theorem 1.1(ii)] prove that on every compact connected Riemannian manifold of dimension at least three with nonempty boundary, there are smooth conformal metrics equal to the original metric on the boundary whose first positive Steklov eigenvalues tend to infinity. In particular, for $A > 0$ choose such a metric k with $\sigma_1(M, k) > 2A$. Applying proposition 2.3 to the inclusion $M \hookrightarrow \mathbb{R}^m$ with $\theta = 1/2$ gives a smooth compact connected embedded image Ω with $\partial\Omega = \Sigma$ and $\sigma_1(\Omega) > A$. No orientability assumption on M enters the argument.

3.2 Surface fillings of an arbitrary embedded curve

Now $d = 1$ and $m \geq 3$. Since Σ is a compact connected smooth one-manifold without boundary, it is a circle. We first construct orientable embedded fillings of every sufficiently large genus.

When $m = 3$, Seifert's construction gives an orientable spanning surface for the knot Σ [7, Section 1, pp. 572–573]. To pass from its polygonal formulation to the smooth original curve, take a sufficiently fine polygon in a normal tubular neighborhood of Σ and smooth its vertices. The resulting curve is a normal graph over Σ, and interpolation of its normal displacement is a smooth isotopy. The finite disks and bands in Seifert's construction can be smoothed. Extending the knot isotopy to the ambient space transports this surface to one whose boundary is exactly the original Σ; see [6, Theorems 4.5.1 and 8.1.3] for tubular neighborhoods and isotopy extension.

For $m > 3$ there is a rank-three linear map $\pi : \mathbb{R}^m \to \mathbb{R}^3$ which embeds Σ. Here is the general-position argument. On the open space of rank-three linear maps, the evaluation

$$(P, x, y) \longmapsto P(x - y), \qquad x, y \in \Sigma, \quad x \neq y,$$

is a submersion, since $x - y \neq 0$. Parametric transversality [6, Theorem 3.2.7] gives a residual set of P for which its restriction to the two-dimensional pair space is transverse to zero in $\mathbb{R}^3$, and therefore has no zero. The same argument on the one-dimensional unit tangent bundle of Σ, using $(P, x, v) \mapsto P(v)$, gives immersion. The intersection of these residual sets is nonempty; a compact injective immersion is an embedding.

Let $\Gamma = \pi(\Sigma)$, and choose a smooth orientable Seifert surface S for Γ in $\mathbb{R}^3$. A linear splitting of π identifies the ambient vector space with $\mathbb{R}^3 \times \mathbb{R}^{m-3}$, in which Σ is the graph of a smooth function $z : \Gamma \to \mathbb{R}^{m-3}$. Extend z over S by multiplying its collar extension by a cutoff which vanishes in the interior. The graph over S of this extension is embedded and has boundary Σ. Inverting the linear splitting gives a smooth orientable filling in the original $\mathbb{R}^m$. The splitting is used only for this topological construction; all induced metrics are taken in the original Euclidean coordinates.

Let S_0 be an orientable filling obtained in this way, of genus g_0. In a ball around an interior disk, disjoint from the rest of the surface and its boundary, replace a smaller disk by a punctured torus joined along a thin annulus. In local ambient coordinates this is the usual embedded connected sum in a three-dimensional slice, available for every $m \geq 3$. It is smooth after rounding the seams, preserves orientability and the outer disk collar, and decreases Euler characteristic by two. Repetition gives an embedded orientable filling M_g of genus g for every $g \geq g_0$, with boundary exactly Σ.

Colbois–Girouard–Raveendran [3, Theorem 7] construct compact surfaces (S_l, k_l) of genus $1+l$, with connected boundary, such that

$$\sigma_1(S_l, k_l)\ \mathrm{length}_{k_l}(\partial S_l) \geq Cl, \tag{9}$$

where $C > 0$ is independent of l. Their construction can be chosen orientable. Section 6.4 starts from a connected four-regular expander graph G_l, placing a genus-zero block with four identical product boundary collars at each vertex and pairing circles along graph edges. Orientation-reversing identifications of paired circles give an orientable closed surface. A maximal tree in G_l is embedded in this surface by joining points in the blocks through the corresponding collars. Its regular neighborhood is a disk; removing it leaves one boundary component. To keep the geometry uniform, the source implements this removal with the four carved piece types in its Figure 5, selected according to the vertex degrees $1, 2, 3, 4$ in the maximal tree. Orientations can be chosen along the tree first, and the remaining full circles glued by orientation-reversing collar isometries. These choices leave the pieces and their metric estimates unchanged; the freedom in gluing and its irrelevance to the spectral conclusion are discussed in [3, Remark 37].

Let $L = \mathrm{length}_{g_{\mathbb{E}}}(\Sigma) > 0$ and $L_l = \mathrm{length}_{k_l}(\partial S_l)$. Rescale the intrinsic metrics by

$$k_l' = (L/L_l)^2 k_l.$$

For a constant $a > 0$, boundary length scales by a and Steklov eigenvalues by a^{-1} under $k \mapsto a^2 k$. Thus

$$\mathrm{length}_{k_l'}(\partial S_l) = L, \qquad \sigma_1(S_l, k_l') \geq Cl/L. \tag{10}$$

For sufficiently large l, the embedded filling M_{1+l} exists and is diffeomorphic to S_l, since both are compact connected orientable surfaces with the same genus and one boundary component [6, Theorem 9.3.7]. The boundary value of the diffeomorphism may be chosen to be an arclength isometry. To see this explicitly, start with a surface diffeomorphism $T : M_{1+l} \to S_l$, choose compatible orientations, and let $b : \Sigma \to \partial S_l$ be an orientation-compatible arclength isometry for k_l'. The discrepancy $(T|_\Sigma)^{-1} \circ b$ is an orientation-preserving circle diffeomorphism. In the arclength coordinate of period L, lift it to a smooth increasing function β with $\beta(s+L) = \beta(s) + L$. The functions

$$\beta_t(s) = (1-t)s + t\beta(s), \qquad 0 \leq t \leq 1,$$

have positive derivative. On a boundary collar use $(s, r) \mapsto (\beta_{\tau(r)}(s), r)$, where $\tau = 1$ near the boundary and $\tau = 0$ near the inner edge. This is a diffeomorphism, extends by identity, and adjusts T to have boundary value b.

Pull k_l' back by this adjusted diffeomorphism to a metric k on the actual embedded filling M_{1+l}. Its tangential boundary metric is exactly the Euclidean metric on Σ. By (10), choose l so that $\sigma_1(M_{1+l}, k) > 2A$. Apply proposition 2.3 to the inclusion of this surface with $\theta = 1/2$. Its smooth embedded image Ω has $\partial\Omega = \Sigma$ and $\sigma_1(\Omega) > A$. This proves the theorem in the remaining dimension and hence for all the stated d, m, Σ and A. □

AI-use statement

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